

3(a)	1 <i>idea of</i> use primers, specific / complementary / AW, to mutant (<i>HBB</i>) allele ; 2 denature DNA / separate DNA into (two) strands ; 3 annealing / binding, of primers ; 4 DNA, elongation / extension / synthesis / replication ;	4
3(b)	homozygous normal = 3 homozygous mutant = 1 heterozygous = 2 and 4 <i>one correct = one mark</i> <i>all three correct = two marks ; ;</i>	2
3(c)	any four from: 1 take DNA sample from cell (of person) ; 2 make DNA single-stranded ; 3 add fluorescent, label / tag / dye / chemical, to the DNA (sample) ; R marker gene 4 microarray / chip, has (ss)DNA, probe(s) for mutant (<i>HBB</i>) allele ; 5 (ss)DNA, hybridises with / binds to / complementary base pairs with, probe(s) (on microarray) ; 6 fluorescence shows presence (of mutant allele) ; 7 AVP ; e.g. <i>ref. to</i> wash off excess (ss)DNA after hybridisation UV light / laser (scan), to see fluorescence	4